Laboratory Work 5

Java OOP and Event Registration

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Purpose: build a Java console application using classes, encapsulation, ArrayList, and input validation.

Program Structure

Registration represents one registration: ID, full name, email, student group, and event title. Its private fields are accessed through getters. The constructor validates the values, and toString() produces readable output.

RegistrationManager owns the ArrayList, assigns IDs, and handles searching, deletion, and filtering. Deletion uses a list index. The ID counter never decreases, so deleted IDs are not reused.

Main displays the menu in a loop, reads input with Scanner.nextLine(), and calls the manager methods. NumberFormatException is handled when converting text to numbers. Invalid input returns the user to the menu.

Running and Testing

Test environment: OpenJDK 21; source code compiled with --release 17. Run with ./gradlew --console=plain run, or launch Main.main() in IntelliJ IDEA. The Gradle build and run tasks completed successfully.

All 18 test scenarios passed when running the compiled Java classes. They cover an empty list, two registrations, trimming, invalid names/emails/groups, nonnumeric input, unknown menu and event choices, searching, filtering, deletion, ID uniqueness, and exit.

Data is stored in memory only. The events are Web Design Workshop, JavaScript Quiz, and IT Career Talk. The application uses no external libraries, web server, or database.

The following images show excerpts from actual console sessions, including typed values and program output. Repeated menu blocks are omitted for readability.

Adding Registrations and Handling Errors

1. Valid Registration and List Output

```
java registration.Main | session 1
Choose an option: 1
Full name:   Kamal Meiirbek
Email:   kamal@example.com
Student group:   IT3 2502DS
1. Web Design Workshop
2. JavaScript Quiz
3. IT Career Talk
Select event: 2
Registration created. ID: 1

Choose an option: 2
1 | Kamal Meiirbek | kamal@example.com | IT3 2502DS | JavaScript Quiz

Choose an option: 0
Goodbye!
```

The registration receives ID 1. Names with spaces are read correctly, and leading and trailing spaces are removed.

2. Short Name and Nonnumeric Input

```
java registration.Main | session 2
Choose an option: 1
Full name: Al
Email: ali@example.com
Student group: IT3 2502DS
1. Web Design Workshop
2. JavaScript Quiz
3. IT Career Talk
Select event: 1
Error: Full name must contain at least 3 characters.

Choose an option: abc
Error: Enter a whole number.

Choose an option: 0
Goodbye!
```

The name Al is rejected. Entering abc does not stop the application: it displays an explanation and returns to the menu.

Searching Filtering and Deletion

3. Existing ID, Missing ID, and Empty Event Filter

```
java registration.Main | session 3
Choose an option: 3
Enter ID: 1
1 | Kamal Meiirbek | kamal@example.com | IT3 2502DS | JavaScript Quiz

Choose an option: 3
Enter ID: 99
Registration not found.

Choose an option: 5
1. Web Design Workshop
2. JavaScript Quiz
3. IT Career Talk
Select event: 1
No registrations for this event.

Choose an option: 0
Goodbye!
```

4. Deleting ID 1 and Creating ID 2

```
java registration.Main | session 4
Choose an option: 4
Enter ID: 1
Registration deleted.

Choose an option: 1
Full name: Ali Karim
Email: ali@example.com
Student group: IT3 2502DS
1. Web Design Workshop
2. JavaScript Quiz
3. IT Career Talk
Select event: 1
Registration created. ID: 2

Choose an option: 2
2 | Ali Karim | ali@example.com | IT3 2502DS | Web Design Workshop

Choose an option: 0
Goodbye!
```

Conclusion

The application consists of three classes with separate responsibilities. Encapsulation protects registration fields, and constructor validation prevents invalid data from being stored. ArrayList supports storing, searching, and deleting registrations. Testing confirmed that input errors are handled and deleted IDs are not reused.